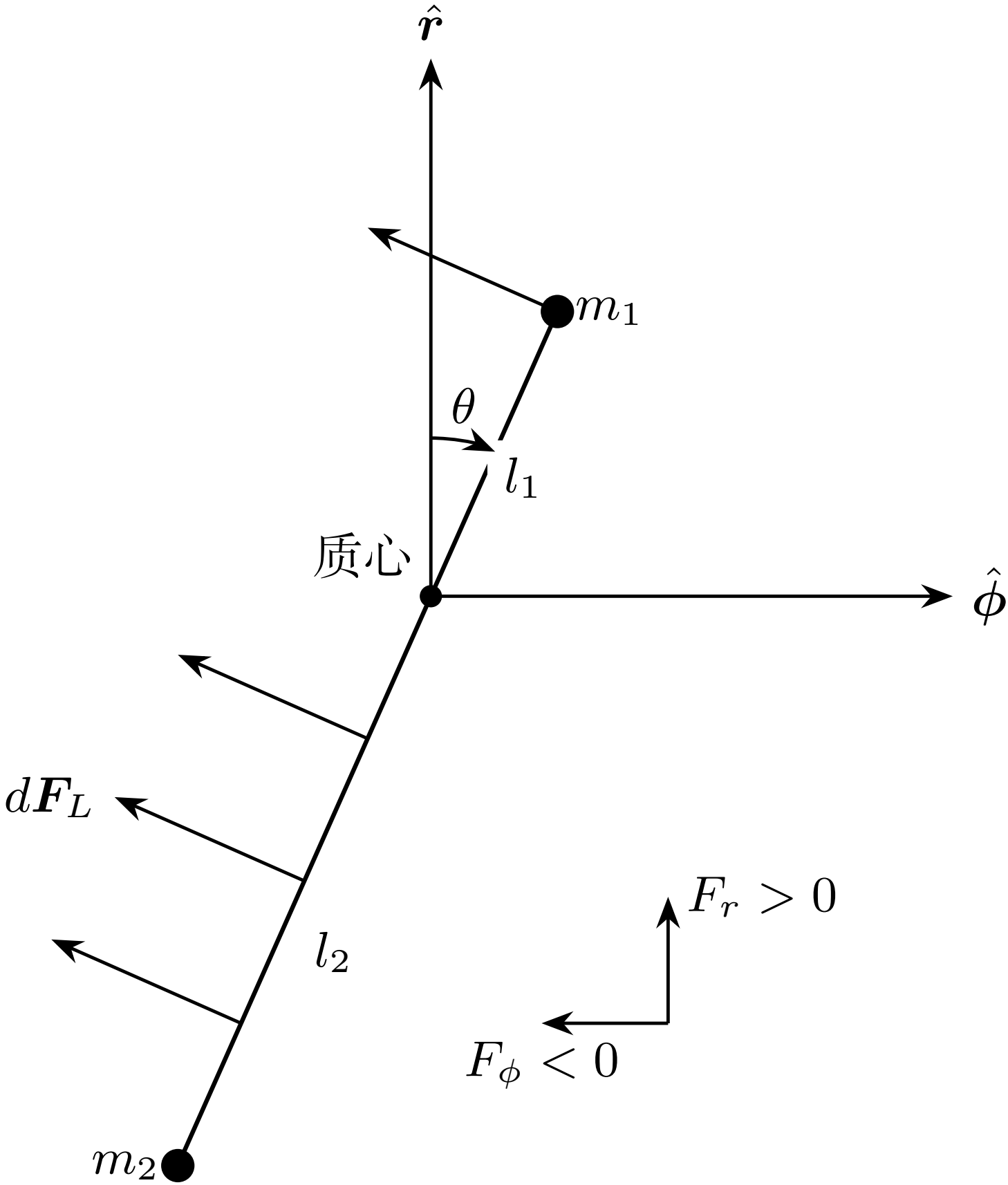
$\hat{\boldsymbol{r}}$
m_1
θ
l_1
质心
$\hat{\boldsymbol{\phi}}$
$d\boldsymbol{F}_L$
l_2
$F_r > 0$
$F_\phi < 0$
m_2
$\tau_{g,z} < 0,\ \tau_{L,z} > 0$